



Instituto Tecnológico Superior del Oriente del Estado de Hidalgo

Título: Reporte Ejecutivo: Métodos Numéricos de Derivación e Integración

Materia: Métodos numéricos

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Grupo / Semestre: EJ26

Código

```
T = 80
t = 0
h = 0.5
tf = 1.5

i = 0

print("i\t t\t T\t\t f(t,T)")

while t <= tf:

    f = -0.1 * (T - 20)

    print(i, "\t", t, "\t", round(T,4), "\t\t", round(f,4))

    if t == tf:
        break

    T = T + h * f
    t = round(t + h, 2)

    i += 1

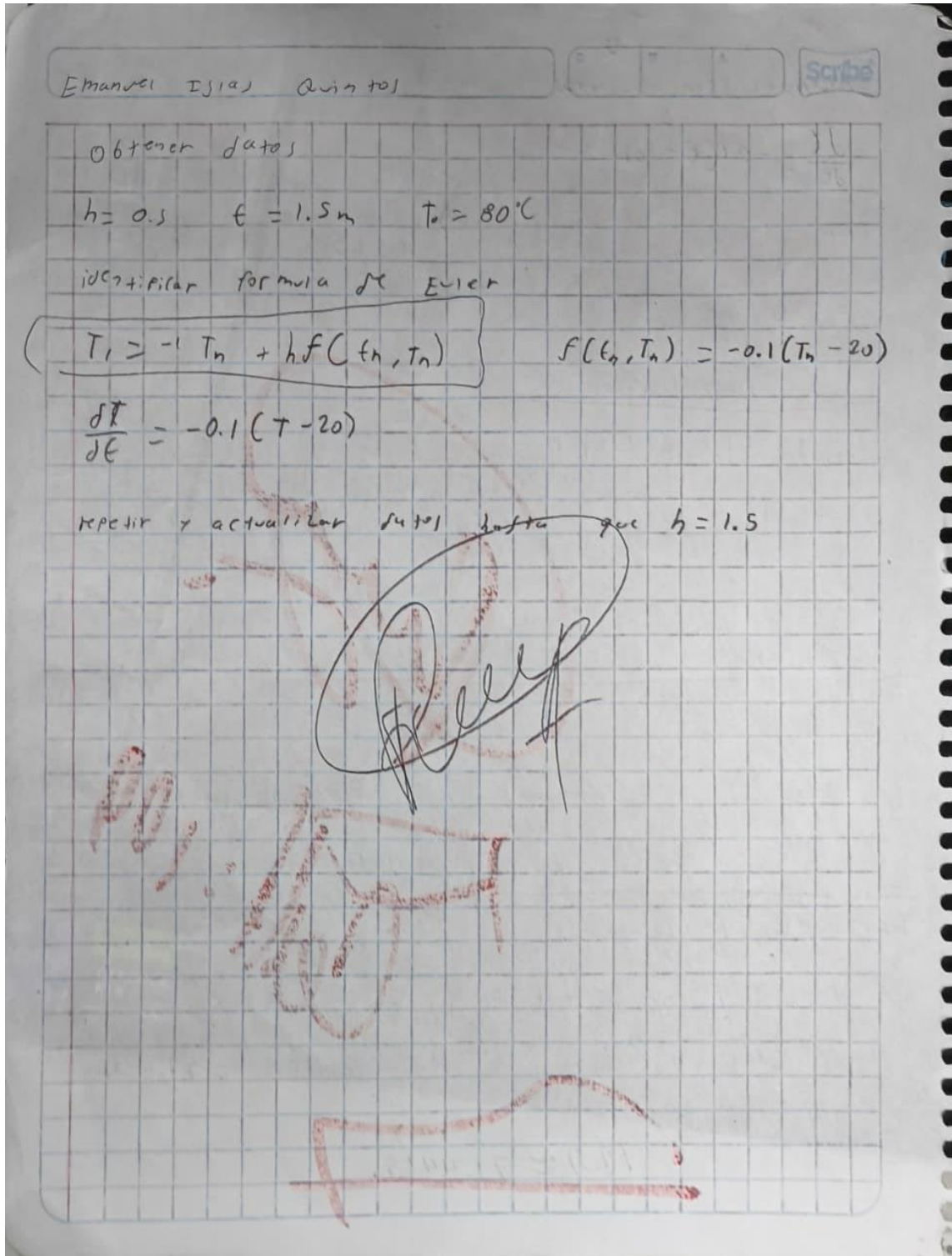
print("\nTemperatura aproximada:")
print("T(, tf, ") = ", round(T,4), "°C")
```

```
PS C:\Escuela\4to\Métodos_numéricos\tema 6\Examen> c
pData\Local\Programs\Python\Python310\python.exe' 'c:
\libs\debugpy\launcher' '56593' '--' 'C:\Escuela\4to\

i      t      T      f(t,T)
0      0      80      -6.0
1      0.5    77.0    -5.7
2      1.0    74.15   -5.415
3      1.5    71.4425  -5.1443

Temperatura aproximada:
T( 1.5 ) = 71.4425 °C
```

Firmas



J6

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$$\frac{dT}{dt} = -0.1(t - 20)$$

$$T(0) = 80^\circ\text{C} \quad h = 0.5 \quad \ell = 1.5 \text{ m}$$

$$t_{n+1} = T_n + h f(t_n, T_n)$$

$$f(t, T) = -0.1(t - 20)$$

It 0

$$T_{n+1} = T_n + 0.5[-0.1(T_n - 20)]$$

$$t_0 = 0$$

$$f(t_0, T_0) = -0.1(80 - 20) = -6$$

$$T_1 = T_0 + h f(t_0, T_0)$$

$$T_1 = 80 + 0.5(-6) = 77 \quad \ell = 0.5$$

It 1

$$t_1 = 0.5 \quad h = 0.5 \quad T_1 = 77$$

$$f(t_1, T_1) = -0.1(77 - 20) = -5.7$$

$$T_2 = 77 + 0.5(-5.7) = 74.15^\circ\text{C} \quad t_2 = 1.0$$

It 2 $t_2 = 1 \quad T_2 = 74.15$

$$f(t_2, T_2) = -0.1(74.15 - 20) = -5.415$$

$$T_3 = 74.15 + 0.5(-5.415) = 74.15 - 2.7075 = 71.4425$$

$$t = 1.5$$

$$T(1.5) \approx 71.4425$$